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Effects of a 12-Week Hybrid Analog–Digital Cognitive Training on Cognitively Unimpaired Older Adults: A Pilot Single-Blind Randomized Controlled Trial

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ABSTRACT

Background and Purpose: Global population aging heightens concern about cognitive decline, underscoring the need for scalable interventions. While computerized cognitive training offers benefits, many seniors face device-related challenges that limit adherence, and facility-based programs pose logistical barriers. To address these, we evaluated a 12-week home-based Hybrid Analog–Digital Cognitive Training (HCT) program, which paired a sensor-embedded, pressure-sensitive paper workbook with a Bluetooth-linked television, to enhance usability and engagement among cognitively unimpaired (CU) older adults.
Methods: In this single-blind, randomized controlled trial, 66 CU adults aged ≥60 years were allocated to HCT or control (each n=33) groups. Participants had literacy and Korean Mini-Mental State Examination scores above −1.5 standard deviation. The HCT group performed 30-minute sessions 5 times weekly for 12 weeks, while controls continued usual activities. Primary outcomes were changes in composite Z-scores for frontal, attention, and memory domains, derived from the Cambridge Neuropsychological Testing Automated Battery. Z-scores were direction-weighted and aggregated by domain. Between-group differences were analyzed using linear mixed-effects models.
Results: Baseline characteristics were similar between groups. After 12 weeks, the HCT group showed a significant group × visit interaction in the frontal domain (estimate, 0.29; standard error, 0.13; p=0.03), indicating improved executive function. No significant differences were found in attention or memory domains. Also, the HCT group exhibited improvements in psychological profiles.
Conclusions: A home-based HCT program improved frontal executive function in CU older adults. The hybrid analog-digital design may help reduce digital barriers, support usability, and foster sustained participation. Larger trials with active controls, modality-specific comparators, and extended follow-up are warranted to confirm benefits and optimize delivery.

Trial Registration: Clinical Research Information Service Identifier: KCT0007922

Keywords: Analog-Digital Conversion; Cognitive Training; Brain Training; Cambridge Neuropsychological Test Automated Battery

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Conflict of Interest

The authors have no financial conflicts of interest.

Author Contributions

Conceptualization: Kim GH; Data curation: Chung S, Moon S; Formal analysis: Chung S, Moon S; Funding acquisition: Kim BR; Investigation: Kim BR; Project administration: Kim BR; Resources: Kim BR, Kim GH; Supervision: Kim GH; Validation: Kim GH; Visualization: Chung S; Writing - original draft: Chung S; Writing - review & editing: Chung S.

INTRODUCTION

The global acceleration of population aging has emerged as a critical public health challenge, with particularly rapid increases observed in Asia and South America, including Korea.[1] It is projected that by 2025 these two regions will account for 75% of the global population aged 65 years and older, with Korea's own population aged ≥65 rising from 7.07 million in 2017 to over 10 million in 2025 and an estimated 19.01 million by 2050.[2] This demographic shift has been accompanied by a growing burden of age-related cognitive decline and dementia, which currently affects more than 55 million individuals worldwide and is expected to triple by 2050, placing unprecedented strain on healthcare systems and caregivers.[3]

This growing burden underscores the urgent need for scalable, accessible, and sustainable interventions to preserve cognitive function and delay cognitive decline in older adults.

Among the various preventive strategies, non-pharmacological interventions, particularly cognitive training programs have gained increasing attention.[4] Traditional paper-based cognitive training programs have long served as standard interventions; however, these approaches can differ in scalability, adaptability, and engagement often pose challenges.[5] To address these limitations, computerized cognitive training (CCT) programs have been developed, offering benefits such as automated difficulty adjustment, standardized delivery, and immediate performance feedback.[6]

Recent meta-analyses indicate that self-guided digital interventions yield small to moderate improvements in executive function, memory, and attention compared to both active and passive controls.[7] However, uptake and engagement with digital interventions varies widely among cognitively unimpaired (CU) older adults, in part due to barriers related to unfamiliar user interfaces, small touchscreen elements, and complex navigation systems, especially those with lower digital literacy or technology self-efficacy.[8,9] Conversely, some older adults prefer traditional, tangible formats such as paper-based workbooks. This diversity in modality preference highlights the need for customizable, user-centered interventions that accommodate individual preferences.[10]

To address these considerations, a Hybrid Analog–Digital Cognitive Training (HCT) program that integrates paper-based and digital modalities may serve as an alternative for older adults. In this context, the HCT program is a home-based intervention that combines a pressure-sensitive paper workbook with a bluetooth-connected television interface, delivering interactive cognitive tasks through a familiar medium and large, easily readable display. This design aims to enhance usability and adherence while preserving the adaptability inherent in digital platforms.

Therefore, this single-center, randomized controlled trial aimed to evaluate whether a 12-week HCT program can elicit measurable improvements in global and domain-specific cognitive functions, particularly executive function, memory, and attention, among CU older adults. Additionally, recognizing that psychological well-being is an essential component of healthy aging, we also investigated the program's impact on anxiety, depression, and quality of life, aiming to assess whether HCT confers comprehensive benefits extending beyond cognition.

METHODS

Study design

This study was a 12-week, prospective, single-blind, randomized controlled trial conducted from October 17, 2022 to March 30, 2023 through public notices at the Yangcheon Dementia Center and Ewha Womans University Mokdong Hospital in Seoul. Participants were randomly allocated in a 1:1 ratio to either the HCT intervention group (n=33) or the control group (n=33).

Participants

A total of 72 older adults aged 60 years or older were initially screened for eligibility. Inclusion criteria included literacy and a Korean version of Mini-Mental State Examination, Second Edition (K-MMSE-2) score within 1.5 standard deviations above the age- and education-adjusted normative mean, consistent with preserved global cognition.

Exclusion criteria comprised a current or prior history of major medical, neurological, or psychiatric illness; diagnosis of dementia or cognitive impairment requiring pharmacological treatment; terminal illness with life expectancy <3 months; acute medical conditions; acute ischemic or hemorrhagic stroke requiring treatment; or illiteracy.

After screening, 6 participants were excluded due to consent withdrawal, and the remaining 66 were enrolled and randomized to the HCT group (n=33) or the control group (n=33) using a simple random sequence (**Fig. 1**).

Written informed consent was obtained from all participants, and the study protocol was approved by the Institutional Review Board of Ewha Womans University Mokdong Hospital

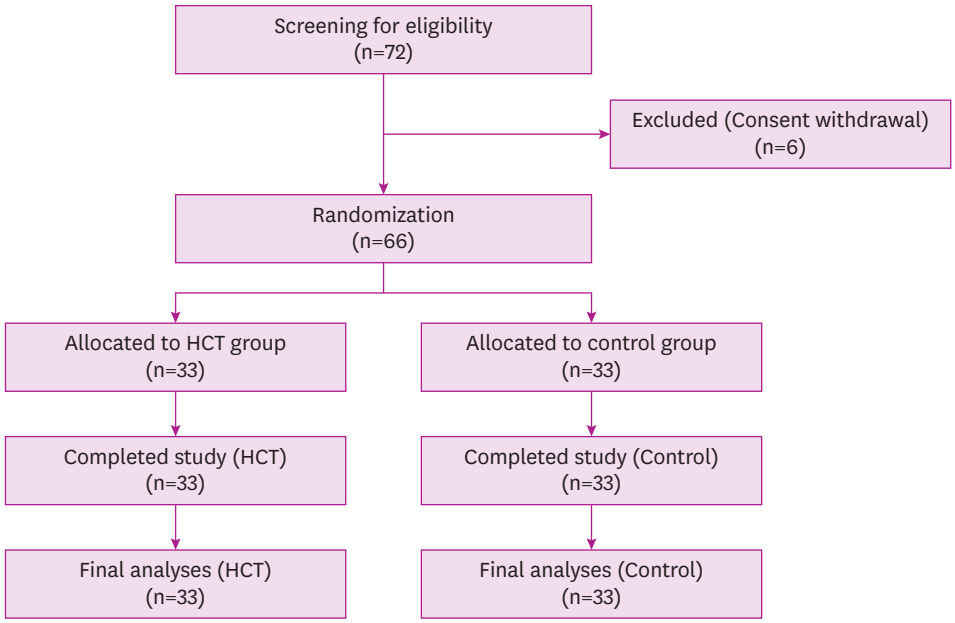

Fig. 1. Study design and flow of the study.
A total of 72 cognitively unimpaired older adults were screened, with 6 excluded due to consent withdrawal. The remaining 66 participants were randomized 1:1 to the HCT group (n=33) and the control group (n=33). The HCT group underwent a 12-week intervention, while the control group maintained their usual activities. Both groups were assessed at baseline and after the 12-week period using the CANTAB cognitive battery.
HCT: Hybrid Analog–Digital Cognitive Training, CANTAB: Cambridge Neuropsychological Testing Automated Battery.

(IRB No. EUMC 2022-08-043-001). This trial was registered at the Clinical Research Information Service (CRIS; KCT0007922).

Intervention: HCT program

The HCT program employed in this study was a commercially available product (FlexTech Co., Ltd., Seoul, Korea) designed to deliver age-appropriate cognitive training programs through a hybrid paper-digital interface. Detailed information on the Silverang software platform is available on the manufacturer's official website (http://www.silverang.net/).

The system consisted of a sensor-embedded workbook ('Silverang') paired with a set-top box that connected to participants' personal televisions, which served as the display device. Because participants used their own televisions, screen size and resolution varied across households. The workbook was a physical paper booklet embedded with pressure sensors rather than a digital tablet. When participants pressed designated areas in response to on-screen instructions, the input was transmitted via the set-top box and displayed in real time on the television screen. The workbook contained 12 semi-permanent pages, each with distinct quiz or game content, and was designed for repeated use with a low failure rate. This configuration allowed large-font, interactive tasks to be displayed in a familiar and easily readable format, thereby enhancing usability among older adults.

The HCT group completed a 12-module, 4-level curriculum explicitly organized around the 3 target domains—frontal/executive, attention, and memory. Frontal training emphasized strategic planning, working memory, and cognitive flexibility (e.g., Brain Games, Visual Search, Problem-Solving Room), attention training targeted sustained/selective attention and processing speed (e.g., Sound Identification, Red Light–Green Light, Whac-a-Mole, Rhythm Game), and memory training focused on episodic/associative memory (e.g., Brain Health Quiz–Memory, My Neighborhood Story), with assessment/feedback and cognitive-screening modules supporting individualized progression (**Supplementary Table 1**).

Participants in the HCT group performed 30-minute sessions 5 times per week for 12 weeks. One session per week was supervised either in person or via remote submission of workbook photographs with the supervision mode determined according to each participant's circumstances and convenience. For remote supervision, participants submitted photographs of completed workbook pages, which were reviewed by the research staff to provide monitoring and feedback.

The control group maintained usual daily activities without structured cognitive intervention. Both groups received standardized general education on dementia prevention.

Adherence in the HCT arm was defined a priori as the percentage of prescribed weekday training days on which the participant completed the daily session during the 12-week intervention, from each participant's program start to end date (weekdays only). Adherence was calculated manually from daily training records compiled in Excel; high adherence was predefined as ≥80%.

Outcome measures

The primary outcomes were changes in domain-specific composite Z-scores of the Cambridge Neuropsychological Test Automated Battery (CANTAB) (comprehensive information on the tests are described at http://www.cambridgecognition.com/academic/

cantabsuite/tests).[11,12] These scores were calculated from seven subtests grouped into three domains: visual memory (Paired Associates Learning [PAL]), Delayed Matching to Sample [DMS], Pattern Recognition Memory [PRM]); executive function (Spatial Working Memory [SWM], One Touch Stockings of Cambridge [OTS]); and attention (Reaction Time [RTI], Rapid Visual Information Processing [RVP]). For each domain, individual subtest Z scores were calculated and averaged to generate the three domain-specific composite scores. All assessments were conducted in a quiet, private room under the direct supervision of the research staff.

A detailed description of the CANTAB variants and composite Z-score calculation procedures can be found in **Supplementary Data 1**.

The secondary outcomes included self-report scales for mood and quality of life: the Subjective Memory Complaints Questionnaire (SMCQ),[13] Geriatric Anxiety Inventory (GAI),[14] Geriatric Depression Scale (GDS),[15] and a standardized quality-of-life questionnaire (WHO-5 Well-Being Index, WB).[16]

Feasibility outcomes included adherence (compliance), summarized as mean ± standard deviation across the HCT arm and as the proportion achieving high adherence (≥80%). Attrition was also recorded.

Statistical analysis

Between-group differences in outcome changes were assessed using linear mixed-effects models with fixed effects for group, visit (baseline vs. week 12), and their interaction (group × visit), adjusting for age, sex, years of education and each individual's baseline value to control for initial group differences. Random intercepts were included for each participant. Analyses were conducted in R (version 4.5.0; R Core Team, Vienna, Austria) using the lmerTest package. The significance threshold for all statistical tests was set at $p<0.05$ (2-tailed).

Primary analyses followed an intention-to-treat approach; a sensitivity analysis restricted to participants achieving high adherence (≥80% of prescribed sessions by manual calculation from daily Excel logs) was prespecified.

RESULTS

Demographic and clinical characteristics

The baseline characteristics of the participants are summarized in **Table 1**.

Table 1. Baseline characteristics of participants

Characteristics	Control (n=33)	HCT (n=33)	*p*-value*
Age (yr)	69.4±4.82	68.2±4.49	0.266
Sex (% female)	22 (66.7%)	26 (78.8%)	0.407
Education (yr)	12.4±3.67	12.2±4.03	0.916
K-MMSE-2	28.7±1.21	28.4±1.41	0.426
CIST	28.3±1.57	28.5±1.64	0.472

Values are presented as number (%) or mean ± standard deviation.
HCT: Hybrid Analog–Digital Cognitive Training, K-MMSE-2: Korean version of Mini-Mental State Examination, Second Edition, CIST: Cognitive Impairment Screening Test.
*$p<0.05$, adjusted by age, sex, and years of education.

No significant between-group differences were observed in age, sex distribution, level of education, or cognitive screening scores (K-MMSE-2 and Cognitive Impairment Screening Test).

At baseline, composite Z-scores of frontal (p=0.071) and memory (p=0.275) domains did not differ significantly between groups, whereas the attention domain showed a significant difference (p=0.038). Among psychological measures, anxiety (GAI: p=0.029), depression (GDS: p=0.026), and psychological well-being (WB: p=0.009) differed significantly between the control and HCT groups, while subjective memory complaints (SMCQ: p=0.184) did not.

Primary outcome measures

At baseline, individual CANTAB subtest Z-scores did not differ between groups (**Supplementary Table 2**). Over 12 weeks, a significant group × visit interaction was observed for the frontal domain (estimate, 0.29; standard error [SE], 0.13; p=0.030), indicating greater improvement in the HCT group than in controls. No significant interactions were observed for attention (estimate, −0.04; SE, 0.10; p=0.724) or memory (estimate, 0.13; SE, 0.09; p=0.160) (**Table 2**, **Fig. 2**). Findings were similar in the per-protocol sensitivity analysis restricted to participants with high adherence (≥80%), which also showed a significant group × visit effect for the frontal domain (estimate, 0.31; SE, 0.13; p=0.023) but not for attention or memory (**Supplementary Table 3**).

Table 2. Changes in composite Z score of 3 domains after 12-week intervention

Domains	Control group (n=33)		HCT group (n=33)		Estimate (SE)	p for interaction*
	Baseline	Follow-up	Baseline	Follow-up		
Attention	0.00±0.55	0.07±0.57	0.30±0.58	0.33±0.48	−0.04 (0.10)	0.724
Frontal	0.00±0.57	0.08±0.60	−0.23±0.42	0.14±0.42	0.29 (0.13)	0.030
Memory	0.00±0.51	0.08±0.46	−0.15±0.59	0.06±0.59	0.13 (0.09)	0.160

Values are described as mean ± standard deviation.
HCT: Hybrid Analog–Digital Cognitive Training, SE: standard error.
*p-values for group-by-visit interaction were calculated using a linear mixed model adjusted for age, sex, years of education and each participant's baseline value.

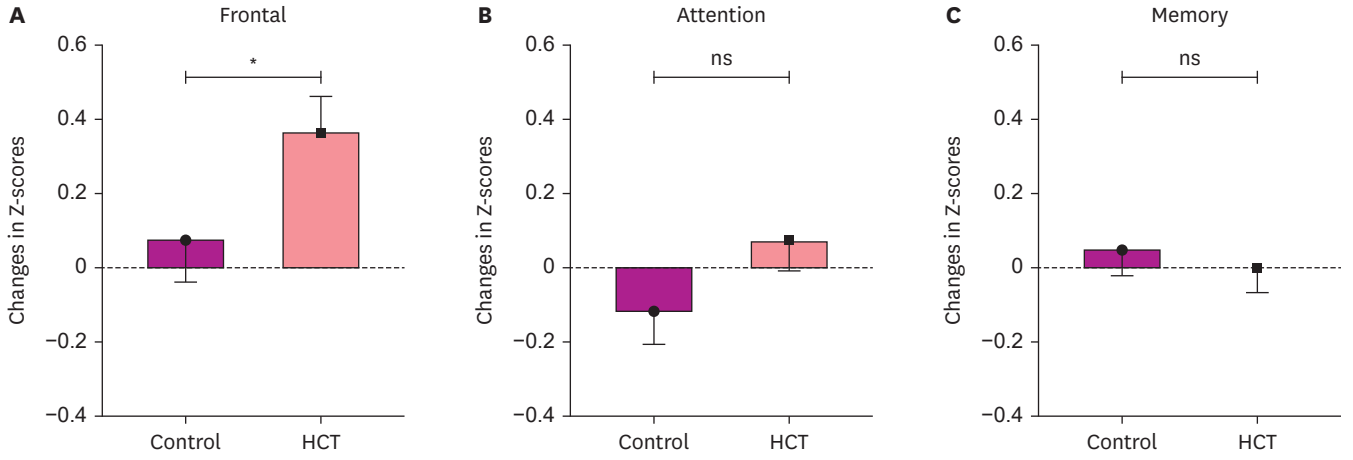

Fig. 2. Changes in domain-specific Z-scores following intervention.
Bar graphs show mean change from baseline to follow-up (± SE) for the control group (n=33, gray) and HCT group (n=33, blue) in (A) Frontal, (B) Attention, and (C) Memory domains. Significance was tested with a linear mixed-effects model for the group × visit interaction; brackets indicate between-group differences in change. Consistent with the annotations, the Frontal domain showed a greater improvement in HCT than control (estimate, 0.29; SE, 0.13; p=0.030), whereas Attention (estimate, −0.04; SE, 0.10; p=0.724) and Memory (estimate, 0.13; SE, 0.09; p=0.160) were not significantly different.
SE: standard error, HCT: Hybrid Analog–Digital Cognitive Training, ns: not significant.
*p<0.05.

Table 3. Changes in secondary outcomes of 3 domains after 12-week intervention

Domains	Control group (n=33)		HCT group (n=33)		Estimate (SE)	*p* for interaction*
	Baseline	Follow-up	Baseline	Follow-up		
GAI	1.5±3.38	2.9±5.19	3.8±4.64	1.8±2.65	−3.30 (1.26)	0.010*
GDS	0.9±2.02	2.2±3.35	2.6±3.67	1.4±2.32	−2.55 (0.89)	0.005*
WB	102.8±13.78	95.5±20.13	93.2±15.12	99.8±15.71	13.91 (5.01)	0.006*
SMCQ	1.9±2.31	3.0±2.98	2.8±3.13	2.0±2.58	−1.88 (0.85)	0.029*

Values are described as mean ± standard deviation.
HCT: Hybrid Analog–Digital Cognitive Training, SE: standard error, GAI: Geriatric Anxiety Inventory, GDS: Geriatric Depression Scale; WB: WHO-5 Well-Being Index, SMCQ: Subjective Memory Complaints Questionnaire.
*p-values for group-by-visit interaction were calculated using a linear mixed model adjusted for age, sex, years of education and each participant's baseline value.

Secondary outcome measures

Compared with controls, the HCT group had greater reductions in anxiety (GAI: estimate, −3.30; SE, 1.26; p=0.010) and depression (GDS: estimate, −2.55; SE, 0.89; p=0.005), and greater improvements in psychological well-being (WB: estimate, +13.91; SE, 5.01; p=0.006). Subjective memory complaints also decreased in the HCT group relative to controls (SMCQ: estimate, −1.88; SE, 0.85; p=0.029), although the effect was borderline (**Table 3**). Results were consistent in the per-protocol analysis (≥80% adherence), which likewise showed significant group × visit effects for anxiety, depression, psychological well-being, and subjective memory complaints (GAI: p=0.007; GDS: p=0.004; WB: p=0.005; SMCQ: p=0.036) (**Supplementary Table 4**).

DISCUSSION

This single-center randomized controlled trial provides preliminary evidence that a 12-week HCT program can produce measurable benefits in both cognitive and psychosocial domains among CU older adults.

In terms of cognitive outcomes, the HCT group demonstrated a significant improvement in frontal domain compared to controls, while no between-group differences were observed in the attention or memory domains. The frontal domain composite mainly reflects strategic planning, cognitive flexibility, and problem-solving.[17] These findings are consistent with prior evidence that cognitive intervention in older adults often produces its earliest and most robust effects in frontal executive function.[18-20] For example, a recent meta-analysis in older adults with subjective cognitive decline reported significant improvements in both executive function and memory, while global cognition remained unchanged.[21] Similarly, an 8-week randomized controlled trial in community-dwelling older adults found that computerized cognitive training significantly improved executive function, and that combining training with preceding aerobic exercise led to broader benefits across multiple executive domains, including set-shifting, inhibitory control, and cognitive flexibility, while memory performance remained unchanged.[22]

Several mechanisms may explain our findings of preferential improvement in frontal executive function. First, many cognitive training programs including the HCT intervention incorporate tasks that directly engage strategic planning, cognitive flexibility, and problem-solving, such as OTS and SWM. These tasks rely heavily on dorsolateral prefrontal and frontoparietal networks, producing domain-specific effects through targeted practice.[23] Second, measurement sensitivity may also contribute. Executive function tasks such as SWM and OTS often exhibit wider performance variability and less ceiling effect than memory tasks

in CU cohorts,[24] making them more sensitive to change within short intervention periods. Taken together, these results suggest that executive processes mediated by the frontal cortex may be particularly responsive to cognitive intervention, even via hybrid analog–digital formats like ours.

In addition to cognitive gains, it is noteworthy that the HCT group demonstrated significant improvements in mood and psychological well-being. These findings align with prior evidence that cognitive interventions can yield modest but meaningful improvements in both cognition and mood, particularly among cognitively intact and functionally independent older adults. For instance, GBD 2019 Dementia Forecasting Collaborators found improved mood alongside cognitive benefits in older adults receiving CCT,[3] and speed-of-processing training using the Road Tour program was associated with reduced depressive symptoms among senior living residents.[25] Of note, participants in the HCT group had higher baseline anxiety and depression scores than controls, making the observed post-intervention improvement in psychological well-being particularly noteworthy. These changes may reflect enhanced self-efficacy, reduced cognitive stress, and increased engagement associated with regular structured training schedule of the HCT program, as shown in other cognitive intervention studies—such as Irigarays et al.,[26] who reported that a structured cognitive training program involving attention, memory, and executive function tasks contributed to improvements in psychological well-being and quality of life among healthy older adults. Similarly, a systematic review by da Silva et al.[27] concluded that cognitive interventions in mature and older adults consistently promote psychological well-being and quality of life.

Engagement with the HCT program was high. Of the 33 participants randomized to the training arm, 31 (94.7%) achieved ≥80% session compliance; the remaining two participants completed 58.3% and 45.0% of sessions without discontinuation, yielding a mean compliance of 92.1% across the training group. This level of adherence exceeds thresholds associated with efficacy in randomized trials of cognitive training in older adults: a recent meta-analysis found that training effects were significant when adherence rates exceeded 80%.[18]

Moreover, adherence in our cohort was comparable to or above the median reported across CCT trials in cognitively healthy older adults (median, 90%; interquartile range, 81–95). Notably, there was no attrition in the HCT arm, supporting the feasibility and acceptability of a hybrid analog–digital format in cognitively unimpaired older adults, in line with recent home-based CCT reports that achieved near-complete session completion and no dropouts. Together with the observed improvements in executive function and psychological well-being, these engagement metrics suggest that HCT is not only effective but also scalable for real-world implementation.

This study has several limitations. It was conducted at a single center with a relatively small sample size, limiting generalizability. The short intervention period and absence of long-term follow-up preclude assessment of the durability of effects. Variation in television screen size and resolution across households may have introduced heterogeneity in task performance, which we were unable to control. Neuroimaging or biomarker data were not obtained, preventing direct evaluation of neural mechanisms underlying cognitive gains. In addition, the HCT group had higher baseline anxiety and depression scores than controls, raising the possibility that part of the observed psychological improvement could reflect regression to the mean, rather than the intervention effect alone. Another limitation of

the current study is the use of a passive control group. Employing an active control condition would have allowed a clearer distinction of the specific effects of HCT from non-specific factors such as expectancy or general engagement. Furthermore, the absence of paper-only or digital-only comparator arms precludes firm conclusions about the hybrid format's relative advantage. Therefore, the findings should be interpreted with caution, and future randomized trials directly comparing hybrid, paper-only, and digital-only interventions will be necessary to delineate their respective benefits.

Nonetheless, despite these limitations, this study provides preliminary randomized controlled evidence on the feasibility and potential benefits of a home-based HCT program in CU older adults. The program produced measurable gains in executive function and notable improvements in psychological well-being, even in a short-term 12-week period. These findings suggest that HCT may represent a practical, scalable, and accessible approach to support both cognitive and emotional health.

SUPPLEMENTARY MATERIALS

Supplementary Data 1

Detailed description and construction of individual CANTAB outcome variables used in the primary analysis

Supplementary Table 1

Curriculum of the HCT program

Supplementary Table 2

Baseline Z-scores of individual CANTAB cognitive variables

Supplementary Table 3

Comparison of composite Z score of 3 domains (ITT vs. PP)

Supplementary Table 4

Comparison of psychological factors of 3 domains (ITT vs. PP)

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